

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2023-24

Paper: CS-105 — Discrete Mathematics

Time: Three hours

Maximum Marks: 70

Instructions: Attempt any five questions including Question No. 1 which is compulsory. All questions carry equal marks.

Question 1. Answer in one word/sentence: [10]

- (a) Write the full form of IP.
- (b) Name the first internet.
- (c) Name one protocol used for file transfer.
- (d) Name the protocol used for transfer over the World Wide Web.
- (e) The malware that can spread automatically over a network is called:
- (f) The malware that locks victim's file and demands money for unlocking is called:
- (g) Give an example of a device that can act both as an input and an output device.
- (h) Can a computer virus spread automatically over the network?
- (i) Processor speed is measured in:
- (j) Network speed is measured in:

Question 2.

- (a) Differentiate between data and information with the help of an example. [5]
- (b) Explain in detail what is data processing along with all the steps that are followed for data processing. [10]

Question 3.

- (a) What are the different types of computer software? Explain with examples. [5]
- (b) What are the different types of digital computers? Explain each of them in detail. Also give examples of each of them. [10]

Question 4.

- (a) Differentiate between compiled and interpreted languages with the help of examples. [5]
- (b) What is an Electronic Mail (E-Mail)? What are its advantages and disadvantages? Give an example of an email client. Explain the basic functionalities of an E-mail Client. [10]

Question 5.

- (a) Explain the use of ICT in Health Care Sector. [5]
- (b) Explain any five different types of computer viruses in detail. [10]

Question 6.

- (a) What is Indexing in the context of a search engine? Explain. [5]
- (b) Explain the IP addresses used for addressing over the Internet. Also differentiate between the IPv4 and IPv6 addressing schemes. Why are two different schemes used? [10]

Question 7.

- (a) What are the different criteria of judging the quality of service provided by a network? Explain each of them briefly. [5]
- (b) Explain the structure of the World Wide Web in detail. [10]